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PAPER_987: WSTP F_U_Bi_i Symbolic Export — Wolfram Language Expression #51

Abstract

Expression #51 in `wstp_kernel_demo_runner.py` encodes the complete 6-layer F_{U,Bi_i} master equation in Wolfram Language for symbolic evaluation via the Wolfram Symbolic Transfer Protocol (WSTP). The export enables closed-form simplification, series expansion, and parametric analysis of all six layers simultaneously within the Mathematica kernel.

1. Wolfram Language Definition

```

Module[{G, Msun, Rsun, AU, SSq, betai, kappa, omegaSCm, ...},
  (* Constants *)
  G = 6.674*^-11; Msun = 1.989*^30; ...

  (* Layer functions *)
  Ug26[M_, r_] := Sum[G*M/r^2 * SSq * i/26, {i, 1, 26}];
  Ub26[M_, r_] := Sum[G*M/r^2 * Exp[-SSq*i/26] * betai, {i, 1, 26}];
  Um[M_, r_] := G*M/r^2 * 1*^-8;
  UA[M_, r_] := -G*M/r^2 * 1*^-10;
  Fn[M_, r_] := G*M/r^2 * 1*^-7;
  Phi[gamma_] := Exp[-(omegaSCm - omegaSCm)^2/(2*gamma^2)] * S26;
  Enet[t_, gamma_] := (2*(Ub26[M,r]/(Ug26[M,r]+0.001))-1)*Exp[kappa*t]*S26;

  (* Master equation *)
  FUBii[M_, r_, t_, gamma_] :=
    Ug26[M,r] + Um[M,r] + UA[M,r] - Ub26[M,r] +
    Fn[M,r] * S26 * Phi[gamma] * Enet[t,gamma];

  (* Evaluate at three radii *)
  Table[FUBii[Msun, R, 86400, 0.1*^12], {R, {Rsun, AU, 10*AU}}]
]

```

2. Symbolic Capabilities

The WSTP export enables: - **Series expansion:** `Series[FUBii[M, r, t, g], {r, Infinity, 3}]` — asymptotic behavior - **Differentiation:** `D[FUBii[M, r, t, g], r]` — force gradient (tidal) - **Integration:** `Integrate[FUBii[M, r, t, g], {r, Rsun, AU}]` — work

done - **Simplification:** FullSimplify[FUBii[M, r, t, g]] — closed-form reduction -
Parametric plots: Plot3D[FUBii[Msun, r, t, 0.1*¹²], {r, 1*⁸, 1*¹²}, {t, 0, 86400}]

3. Integration with WSTP Pipeline

Expression #51 joins the existing 50 WSTP expressions. The WSTP kernel demo runner dispatches it via:

```
link.putFunction("EvaluatePacket", 1)
link.put(expr_51_code)
link.endPacket()
result = link.getResult()
```

4. Implementation

Added as expression #51 in wstp_kernel_demo_runner.py, function_build_expressions().
Total WSTP expressions: 51.

References

- PAPER_979: Complete 6-Layer F_U_Bi_i
 - PAPER_985: Production Kernel
-

§A. Cosmogenesis-Linked Lagrangian

The Wolfram export allows symbolic computation of $\delta S/\delta\phi$ directly in Mathematica, cross-validating the numerical Euler-Lagrange result from PAPER_981.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** Wolfram's Sum and Exp handle the vacuum density series analytically.
- **DVP:** Symbolic differentiation reveals the DPM dipole contribution at each order.
- **BSH:** The buoyancy harmonic sum Sum[Exp[-SSq*i/26]*betai, {i,1,26}] evaluates to closed form via geometric series.